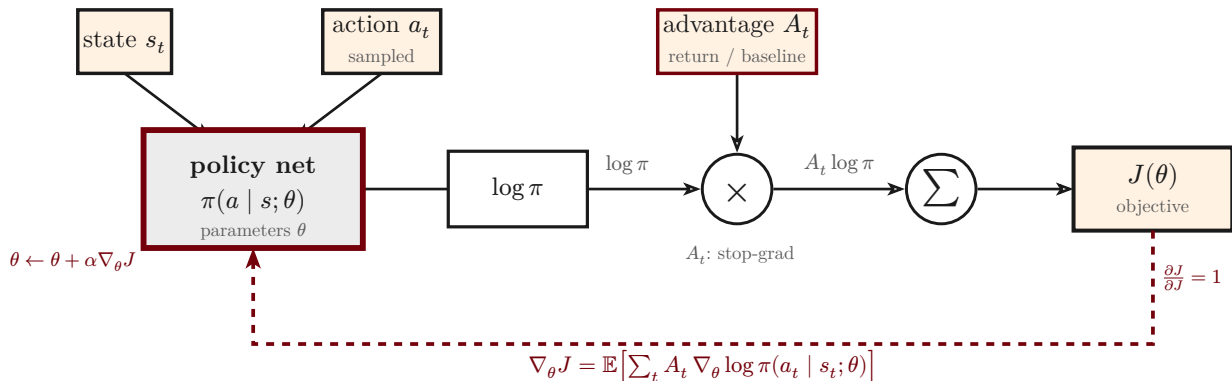


Policy gradient (REINFORCE): the score-function estimator

$$J(\theta) = \mathbb{E} \left[\sum_t A_t \cdot \log \pi(a_t \mid s_t; \theta) \right], \quad \nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_t A_t \cdot \nabla_{\theta} \log \pi(a_t \mid s_t; \theta) \right]$$



————→ forward (sample, score, weight, sum)

- - - - - ➔ backward (policy gradient, ascent on θ)

The estimator weights each action's log-probability by its advantage A_t and sums over the trajectory. Differentiating the surrogate $J(\theta)$ — with A_t held constant (a stop-gradient) — yields the **score function** $A_t \nabla_{\theta} \log \pi$: ascending it raises the probability of actions with positive advantage and lowers it for negative ones.